

通过从未知方向观察平面实现灵活的相机标定

课程作业译稿
原论文作者: Zhengyou Zhang

摘要

本节为 LaTeX 项目骨架占位内容, 待摘要翻译 worker 按用户提供的原始论文补充。

关键词: 相机标定; 内参数; 镜头畸变; 基于平面的灵活标定; 运动分析; 模型获取

1 动机

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

2 基本方程

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

3 相机标定求解

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

4 退化配置

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

5 实验结果

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

6 结论

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

A 模型平面与图像之间单应性的估计

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

B 从矩阵 B 提取内参数

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

C 用旋转矩阵逼近 3×3 矩阵

本节为 LaTeX 项目骨架占位内容, 待翻译 worker 按用户提供的原始论文补充。

参考文献

- [1] S. Bougnoux, "From projective to euclidean space under any practical situation, a criticism of self-calibration," in *Proceedings of the 6th International Conference on Computer Vision*, Jan. 1998, pp. 790–796.
- [2] D. C. Brown, "Close-range camera calibration," *Photogrammetric Engineering*, vol. 37, no. 8, pp. 855–866, 1971.
- [3] B. Caprile and V. Torre, "Using vanishing points for camera calibration," *The International Journal of Computer Vision*, vol. 4, no. 2, pp. 127–140, Mar. 1990.
- [4] W. Faig, "Calibration of close-range photogrammetry systems: Mathematical formulation," *Photogrammetric Engineering and Remote Sensing*, vol. 41, no. 12, pp. 1479–1486, 1975.
- [5] O. Faugeras, *Three-Dimensional Computer Vision: A Geometric Viewpoint*. MIT Press, 1993.
- [6] O. Faugeras, T. Luong, and S. Maybank, "Camera self-calibration: Theory and experiments," in *Proceedings of the 2nd European Conference on Computer Vision*, May 1992, pp. 321–334.
- [7] O. Faugeras and G. Toscani, "The calibration problem for stereo," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Miami Beach, FL, Jun. 1986, pp. 15–20.
- [8] S. Ganapathy, "Decomposition of transformation matrices for robot vision," *Pattern Recognition Letters*, vol. 2, pp. 401–412, Dec. 1984.
- [9] D. Gennery, "Stereo-camera calibration," in *Proceedings of the 10th Image Understanding Workshop*, 1979, pp. 101–108.
- [10] R. Hartley, "Self-calibration from multiple views with a rotating camera," in *Proceedings of the 3rd European Conference on Computer Vision*, Stockholm, Sweden, May 1994, pp. 471–478.
- [11] ———, "In defence of the 8-point algorithm," in *Proceedings of the 5th International Conference on Computer Vision*, Boston, MA, Jun. 1995, pp. 1064–1070.
- [12] R. I. Hartley, "An algorithm for self calibration from several views," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Seattle, WA, Jun. 1994, pp. 908–912.
- [13] D. Liebowitz and A. Zisserman, "Metric rectification for perspective images of planes," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Santa Barbara, CA, Jun. 1998, pp. 482–488.
- [14] Q.-T. Luong and O. Faugeras, "Self-calibration of a moving camera from point correspondences and fundamental matrices," *The International Journal of Computer Vision*, vol. 22, no. 3, pp. 261–289, 1997.
- [15] S. J. Maybank and O. D. Faugeras, "A theory of self-calibration of a moving camera," *The International Journal of Computer Vision*, vol. 8, no. 2, pp. 123–152, Aug. 1992.
- [16] J. More, "The levenberg-marquardt algorithm, implementation and theory," in *Numerical Analysis*, ser. Lecture Notes in Mathematics, G. A. Watson, Ed. Springer-Verlag, 1977, vol. 630.
- [17] C. C. Slama, Ed., *Manual of Photogrammetry*, 4th ed. American Society of Photogrammetry, 1980.
- [18] G. Stein, "Accurate internal camera calibration using rotation, with analysis of sources of error," in *Proceedings of the 5th International Conference on Computer Vision*, Cambridge, MA, Jun. 1995, pp. 230–236.
- [19] B. Triggs, "Autocalibration from planar scenes," in *Proceedings of the 5th European Conference on Computer Vision*, Freiburg, Germany, 1998, pp. 89–105.
- [20] R. Y. Tsai, "A versatile camera calibration technique for high-accuracy 3d machine vision metrology using off-the-shelf tv cameras and lenses," *IEEE Journal of Robotics and Automation*, vol. 3, no. 4, pp. 323–344, Aug. 1987.
- [21] G. Wei and S. Ma, "A complete two-plane camera calibration method and experimental comparisons," in *Proceedings of the Fourth International Conference on Computer Vision*, Berlin, Germany, May 1993, pp. 439–446.

- [22] —, “Implicit and explicit camera calibration: Theory and experiments,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 16, no. 5, pp. 469–480, 1994.
- [23] J. Weng, P. Cohen, and M. Herniou, “Camera calibration with distortion models and accuracy evaluation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 14, no. 10, pp. 965–980, Oct. 1992.
- [24] Z. Zhang, “A flexible new technique for camera calibration,” Microsoft Research, Tech. Rep. MSR-TR-98-71, Dec. 1998. [Online]. Available: <http://research.microsoft.com/~zhang/Calib/>